

442OOLS // META-VISION

OpenCV AI Competition 2026 — Testing & Evaluation Guide

OpenCV 5.0 / 4.10

AWS Bedrock & Graviton

YOLOv8 + ByteTrack

FastAPI Studio UI

1. System Setup & Environment Installation

Clone the code repository and install the pinned dependencies within a Python virtual environment:

```
git clone https://github.com/Insight14/442ools.git
cd 442ools
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

2. Launch Interactive Meta-Vision Web Studio

Launch the FastAPI local backend server and open the browser interface to test the 2D Tactical Radar, Dribble Biomechanics HUD, and Live Telemetry stream:

```
python3 server.py
# Open your web browser at:
# http://localhost:8000
```

3. CLI Direct Computer Vision Pipelines (Standalone Testing)

Execute the core OpenCV 5 analysis scripts directly via terminal CLI:

```
# A. Full Match Tactics, Voronoi Space Hulls & Expected Threat (xT)
python3 src/detect_track.py \
--source input_videos/mbappe_clip.mp4 \
--output output_videos/test_tactics_analyzed.mp4 \
--homography pitch_calibration/homography_mbappe_clip.json

# B. 1v1 Dribble Biomechanics & Posture Equilibrium Tracking
python3 src/dribble_pose_engine.py \
--video input_videos/doku_dribble2.mp4 \
--output output_videos/test_dribble_analyzed.mp4
```

4. AWS Bedrock Agentic Vision Loop Verification

Verify the perception-decision-action loop connecting OpenCV 5 spatial telemetry with Amazon Bedrock tactical reasoning:

```
curl -X POST http://localhost:8000/api/agentic-vision \
-H "Content-Type: application/json" \
-d '{"carrier_id": 10, "defender_dist": 2.1, "pass_prob": 0.78, "delta_xt": 0.052}'
```

5. Verification Checklist & Expected Outputs

Component	Target Script	Expected Result
Tactics Studio	server.py	Live Web UI at localhost:8000
Match Tracking	src/detect_track.py	Bounding boxes, pitch radar, pass lanes
1v1 Biomechanics	src/dribble_pose_engine.py	Skeleton, trunk lean angle, balance score
Agentic AI Loop	src/aws_agentic_coach.py	Automated tactical decision JSON response